G SARAN KASHYAP

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Career Objective

To be part of a reputed organization where I can apply my problem-solving skills, creativity, and technical knowledge to contribute to impactful projects, support organizational goals, and grow professionally.

Education

VNR Vignana Jyothi Institute of Engineering and Technology, Sept 2023 – Present
B.Tech in **Electronics and Instrumentation Engineering**
CGPA: 9.29/10

B.Tech Minor in **Artificial Intelligence and Machine Learning** Aug 2025 – Present
CGPA: 10/10

Narayana Junior College – MPC – May 2023
97.2%

Narayana High School – June 2021
CGPA: 10/10

Work Experience

Milieu Global Pvt. Ltd., Embedded Engineer Intern January 2026 - Present

- Developed an **IoT-based multi-gas monitoring system** using ESP32 with multi-node LoRa communication and GSM-based cloud connectivity.
- Designed a **3-node LoRa architecture** with a relay node to extend communication range and enable reliable data transmission in constrained environments.
- Implemented **MQTT-based data publishing** over GSM (SIM800) for real-time remote monitoring using a cloud broker (HiveMQ).
- Built a **WiFi AP-based web dashboard** for system monitoring and runtime configuration of gas thresholds and calibration parameters.
- Implemented **OTA-style configuration updates** for real-time threshold and calibration management using persistent NVS (Preferences) storage.
- Integrated multiple sensors using **I2C, SPI, and ADC** with I2C multiplexer-based channel selection.
- Developed a **threshold-based alert system** with LED indicators, buzzer alerts, OLED display logging, and SMS notifications.
- Performed **hardware debugging and system optimization**, resolving LoRa frequency mismatch and GSM power stability issues.
- Assisted in **PCB design and schematic development** for multi-sensor integration using EasyEDA.

Hasol Energy Pvt Ltd, Product Engineer Intern June 2025 – September 2025

- Developed **firmware for a dual-axis solar tracker** with focus on **PWM-based motor control**, real-time monitoring, and performance optimization.
- Wrote embedded software enabling seamless **hardware–software integration** and refined control algorithms based on field performance data analysis.
- Analyzed performance data and identified optimization opportunities for improving tracking accuracy and system reliability
- Collaborated with a multidisciplinary team to **design, validate, and field-test** functional prototype models for deployment.

Projects

Fall Detection & Smart Helmet System

- Designed a wearable fall detection and smart helmet system for rider safety and health monitoring
- Implemented fall detection algorithm using ADXL345 accelerometer with acceleration threshold and pattern analysis; monitored vital signs via MAX30100 pulse oximeter

- Integrated Twilio API to send real-time SMS alerts during emergencies
- Built a basic interface to display live sensor data for monitoring

IoT-Controlled Car

- Built a car prototype controlled via IoT using ESP32 and the Blynk mobile app
- Integrated Xbox controller to enable both mobile and joystick-based operation
- Implemented real-time control of movement, speed, and direction
- Demonstrated reliable wireless control for real-time operation
- Gained hands-on experience in hardware–software interfacing

Disease Prediction System

- Built a disease prediction model using **Logistic Regression** (97.6% accuracy) and **KNN** (92.8% accuracy), evaluated across 7 disease classes including dengue, malaria, and typhoid.
- Developed a web interface for real-time symptom input and prediction output.
- Used **Joblib** for model serialization and deployment.

Anti-lock Braking System (ABS) Simulation

- Developed a Bang-Bang controller based ABS simulation model in MATLAB Simulink
- Simulated tire dynamics including slip ratio and braking behavior
- Tested performance under different road conditions
- Analyzed braking performance and stopping distance

Study Group Connector

- Developed a basic platform to connect students for academic collaboration
- Implemented user authentication for secure login and registration
- Built CRUD functionality for managing study groups
- Enabled interaction between users within groups

Technical Skills

Embedded & IoT: ESP32, STM32, Arduino, RTOS, Linux, Firmware development , PCB Design
Communication: UART, I2C, SPI, LoRa, RF, GSM ,MQTT,WIFI
Languages: C, Python, JavaScript, Embedded C
Web Development: HTML, CSS, Node.js, Express (Basic), MongoDB, RESTful APIs
AI/ML: Machine Learning algorithms, Feature Engineering, Model Evaluation
Tools: Arduino IDE, VS Code, GitHub, MATLAB, EasyEDA
Core Competencies: Data Structures and Algorithms, Problem-solving, Hardware Debugging, System Integration

Achievements and Participation

Co-Curricular

- Secured **3rd place** in National Science Day 2025 at VNRVJIET
- Secured **8th place** out of 50 teams in 24-Hour Hackathon **“Innovathon”**
- Participated in 24hr hackathon "**Emerge**"
- Attended IoT workshop **“Laser”** conducted by ISOI
- Won several national-level Olympiads during schooling

Extra-Curricular

- Assisted with stall setup and participant guidance at INCC 6.0
- Assisted AIRBOTS club in organizing events during Convergence 2K25

Certifications

NPTEL – Real Time Operating System
NPTEL – A Brief Introduction of Micro-Sensors
Maven Silicon – Embedded System Design using C Programming